

NYC Taxi Urban Mobility Explorer

Team 11

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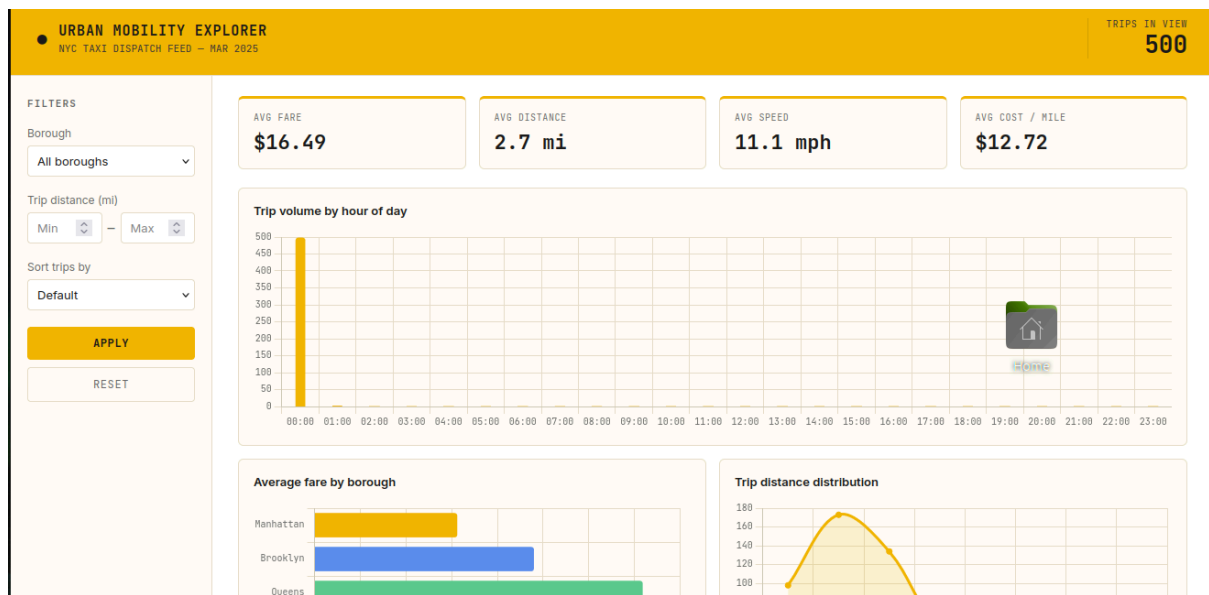
Nshuti Lydivine

The frontend was built using HTML, CSS, and JavaScript. This is a part of the system that the users see and interact with directly in the browser. When you open the dashboard you will see four things, A sidebar on the left where you can filter trips by area or distance. Four boxes at the top showing average fare, distance, speed, and cost per mile. The charts in the middle show patterns in the taxi data, and the table at the bottom show individual trips.

For us to keep things simple and easy to understand, we split the js into four separate files. The first file called [mockData.js](#) creates fake but realistic taxi trip data so that dashboard has something to show before the real data is connected, the second file is called [api.js](#) it's the only place that gets the data, this means if we want to switch from fake data to real data we only change this one file, the third is called [charts.js](#) this draws all the charts on the page, and the fourth file is called [main.js](#) this runs everything, when you click apply it reads your filters and update the charts and tables.

For the charts I used a tool called chart.js , we chose it because it is simple to use, works straight in the browser and makes clean looking charts without needing anything complicated. The colors are yellow and white because I mean when anyone hears a taxi those are the main colors that come to mind, so it makes the dashboard feel connected to what it is showing.

Figure 1: Full dashboard showing all four charts and the filter sidebar



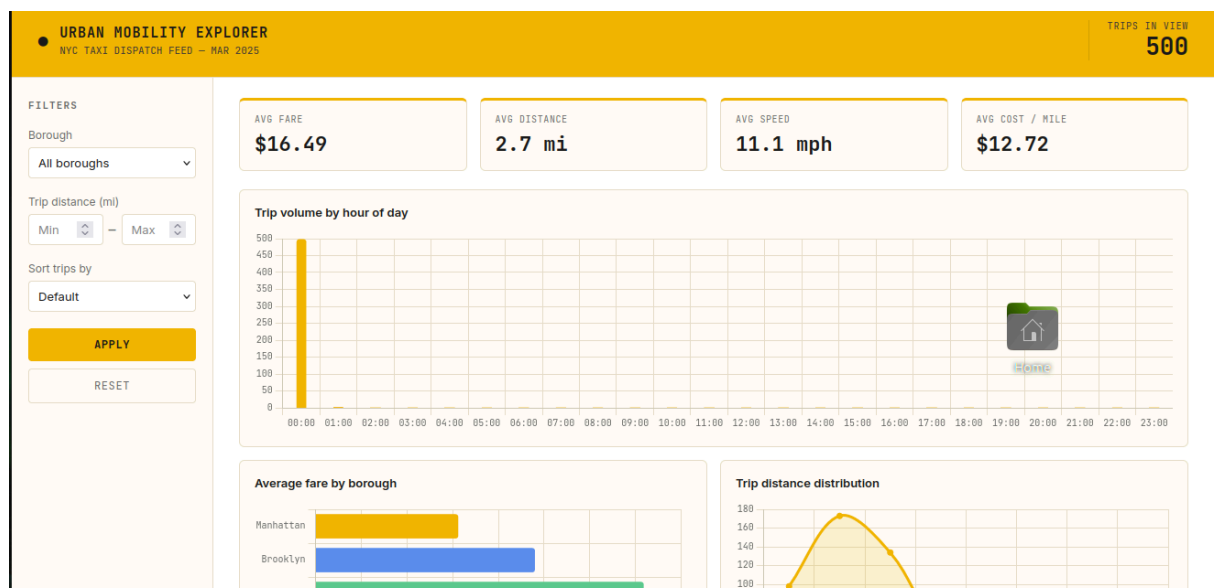
What we found in the Data

After exploring the dashboard we found three interesting things:

- **Most trips happen midnight of New Year's**

As you can see in the trip volume chart, the highest number of trips happened at midnight. This is because the data we loaded is from the 1st Jan 2025, this explains the unusual spike at midnight instead of the typical morning rush hour pattern we would see on a normal day.

Figure 2: Trip volume spike at midnight on New Year's Day



- **Queens has the highest average fare compared to other boroughs**

The average fare by borough charts shows that Queens trips cost about \$38 on average, which is more than Manhattan average of \$16, and Manhattan is the busiest area. But it makes sense when you see where Queens trip starts from.

Figure 3: Average fare by borough



- **Queens trips are dominated by airport pickups**

When we filtered the dashboard to show the Queens borough, the top pickup zones were JFK and LaGuardia Airport, they completely dominated all other zones. And this also explains why Queens has the highest fare, because of the long distances and expensive trips. And the trip table also confirms this, most Queens trips start at JFK or LaGuardia and travel many miles across the city.

Figure 4: Queens filter applied — showing avg fare and mi avg distance

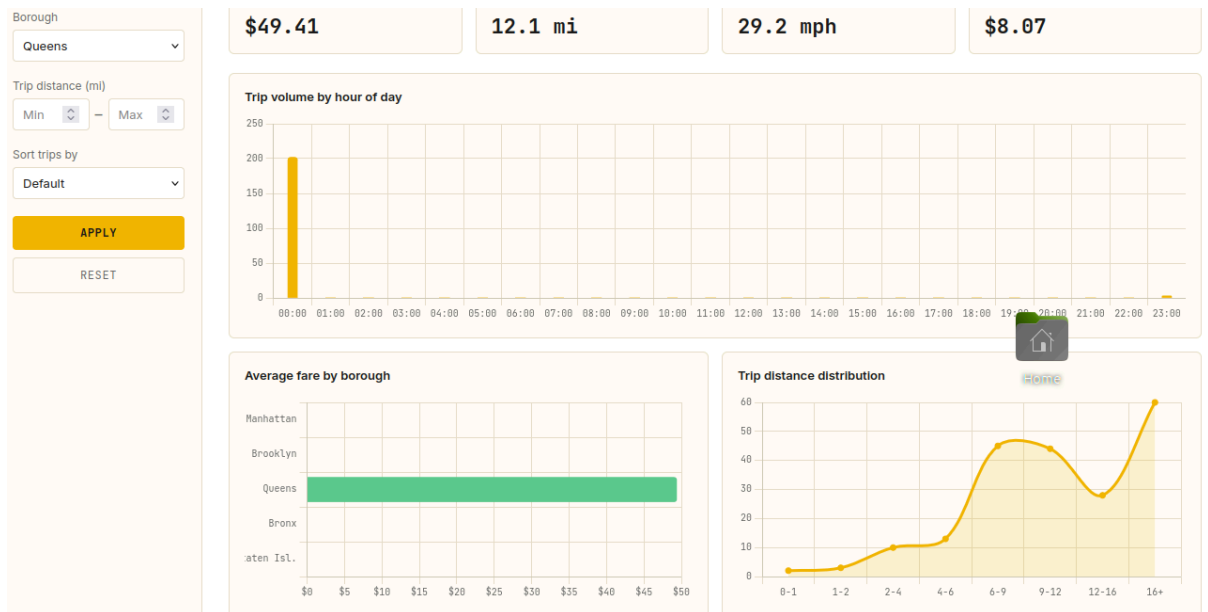


Figure 5: Top pickup zones in Queens



Reflection and Future Work

Honestly this project was fun working on, one of the challenges I faced was that I had to build the dashboard before my teammates finished the backend, so that I would try to manage my time. So I had to find a way to make it work using fake data and then later connect it to the real data. This taught me how to plan ahead and write code in a way that is easy to update. And for the future works if I was to keep building this I would add more filters, make the charts update in real time without clicking apply.